

Durability evaluation of moderate temperature cured E-glass/vinylester systems

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Abstract

FRP composites using vinylester resins are increasingly being fabricated for use in civil and marine structures using processes that incorporate ambient and moderate temperature cure regimes. This often results in the final use of the material when cure progression has not reached one hundred percent. This paper presents results of an investigation into deterioration in pultruded E-glass/vinylester composites due to immersion in deionized water and alkaline solution. Degradation is accelerated through the use of elevated temperatures and results are characterized through moisture uptake measurements, mechanical tests (tension and short-beam-shear), dynamic mechanical thermal analysis (DMTA), and microscopy. It is shown that degradation due to alkali exposure is more severe than due to the deionized water. Data from accelerated tests is used to provide long-term predictions, which are shown to correlate well with experiments in the short-term. The use of DMTA characteristics as a basis for the development of an understanding of degradation phenomena and for further modeling is shown to be a promising avenue for further research.

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1. Introduction

Due to their corrosion resistance, relatively high ultimate failure strains, and damage tolerance, E-glass/vinylester composites are increasingly being considered for use in cost-sensitive primary structural components in marine, offshore and civil infrastructure applications. Beyond their use in hulls, decks (ship, pier, and bridge related) and other replacement structural components, these materials are also used as reinforcement (rebar) in concrete, and for external strengthening of deteriorating or under-strength concrete structures. Due to exigencies of cost and even structural size, these systems are processed using non-autoclave cure processes such as resin infusion, pultrusion, and even wet layup. In a majority of these cases extended service lives, under a harsh and changing environment, are required with minimal maintenance. Although significant research has been conducted on the hygrothermal ageing of composites, most investigations to date have focused on carbon reinforcement, elevated temperature cure regimes such

as in autoclave processing and epoxy resin systems. There is a much lower level of understanding of mechanisms of ageing and degradation in ambient and moderate temperature cure regimes, especially as related to vinylesters [1–3]. It is important to note that E-glass fibers are themselves extremely susceptible to degradation in water and moist environments and the vulnerability of these fibers to deteriorate when in contact with alkalis, such as pore water in concrete which has a high pH, is well established [4–6].

Vinylesters are comprised of low molecular weight polyhydroxyether chains with reactive groups at chain ends. Styrene in the monomeric form is used as a diluent in the resin and an increase in styrene content results in an intrinsic increase in hydrophobicity, volumetric shrinkage and the potential for incomplete polymerization. This leads to a competition between the effects of a slow progression of cure, resulting in an increase in performance attributes, and degradation at the resin and interface levels due to moisture ingress. In addition, the hydrolysis of ester groups can result in the formation of carboxyl groups, resulting in further deterioration due to autocatalysis [7]. It should also be noted that incomplete polymerization not only leads to changes in properties with time, but also induces lower heat

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stability, lower resistance to hydrolysis, and a greater susceptibility for swelling when exposed to solvents [8].

Since the applications under consideration all involve the exposure of the composite to moisture and/or alkali (as related to concrete), it is critical that a mechanistic understanding of materials response be developed at the fundamental level while simultaneously allowing for the development of models not just for the prediction of property changes with time but also for determination of remaining service-life of a structure. This paper presents results of an ongoing investigation into the durability of pultruded E-glass/vinylester composites exposed to a series of controlled environments in an attempt to understand the primary mechanisms and rates of degradation, prior to considering complex regimes such as of indeterminate changes in humidity, temperature cycles, and exposure to other solvents, as would be expected in the field.

2. Materials and test methodology

The E-glass fiber used in this investigation was in the form of Hybon 2001 rovings of 23 m/kN (112 linear yards/lb) yield (manufactured by PPG Industries¹), with typical performance attributes at the fiber level of 72.40 GPa, 1.45 GPa, and 5.04 $\mu\text{m}/\text{m}/^\circ\text{C}$ for tensile modulus, tensile strength, and coefficient of thermal expansion, respectively. The resin system used was a Dow Derakane 441–400 vinylester with a viscosity of 400 cps at 23 $^\circ\text{C}$ and a styrene content of 33% by weight. The resin was chosen for its improved heat resistance (heat distortion temperature of 115–122 $^\circ\text{C}$), enhanced elongation (5–7%), good corrosion resistance and reduced emissions. Based on tests of unreinforced resin plates tensile strengths and moduli were determined to be 82–90 MPa and 3.1–3.5 GPa, respectively.

The unidirectional composites were pultruded in 1.6 mm thick strips of 1524 mm width. Fiber loading, assessed through burn-off procedures, was determined to be approximately 62%. All specimens were preconditioned at 23 $^\circ\text{C}$ and 46% relative humidity prior to initiation of the test program.

In order to adequately assess changes in characteristics of the composite as well as mechanisms of degradation as a result of water sorption, the material was subjected to mechanical testing through tension (ASTM D3039) and short-beam-shear (ASTM D2344) modes. Dynamic mechanical thermal analysis (DMTA) was conducted at a frequency of 1 Hz between 25 and 200 $^\circ\text{C}$. Inductively coupled plasma (ICP) analysis was also conducted on solutions in which the material was im-

mersed in order to assess leachate elements, and fourier transform infrared (FTIR) spectroscopy was used to assess changes in the polymer itself. Moisture uptake was assessed through gravimetric means and both optical and scanning electron microscopy (SEM) were used to study degradation. Specimens were exposed to the following three sets of environments for periods of time up to 75 weeks.

- (a) Set A: Storage at 23 $^\circ\text{C}$ and 46% relative humidity RH, taken to be representative of ambient, “unexposed” conditions
- (b) Set B: Immersion in deionized water
 - (i) at 23 $^\circ\text{C}$
 - (ii) at 40 $^\circ\text{C}$
 - (iii) at 60 $^\circ\text{C}$
 - (iv) at 80 $^\circ\text{C}$
- (c) Set C: Immersion in alkaline solutions formed by using 10.89 g/l of CaCO_3 and 5.95 g/l of $\text{Ca}(\text{OH})_2$ in deionized water to provide a pH of 11.5, and to simulate concrete pore water with a focus on Ca ions
 - (i) at 23 $^\circ\text{C}$
 - (ii) at 40 $^\circ\text{C}$
 - (iii) at 60 $^\circ\text{C}$
 - (iv) at 80 $^\circ\text{C}$

In order to assess reversibility of moisture sorption effects, tensile and short-beam-shear tests were also conducted on specimens that were reconditioned at 23 $^\circ\text{C}$ and 46% RH for 28 days (equal to the original period of conditioning) after removal from the immersion environments.

3. Results and discussion

For clarity in discussion results are presented as related to the method of the characterization used. Wherever possible, however, effects are cross-referenced.

3.1. Moisture uptake

Mass uptake in the specimens immersed in the different solutions showed response that could best be characterized as Fickian although minor variations were seen, especially at longer time periods at 23 $^\circ\text{C}$ when a short period of mass loss due to hydrolysis and associated degradation was noted followed by regain. In all cases, however, rates of uptake and moisture content were seen to increase with temperature. Overall results in terms of maximum percentage weight gain, over the 75-week period of investigation, and diffusion coefficients determined from the mass uptake curves are listed in Table 1.

A comparison with the values of percentage weight gain and diffusion coefficients in Table 1 shows that both

¹ Commercial designations provided in this paper are for purposes of identification only.

Table 1
Characteristics of mass uptake

Temperature (°C)	Immersion in alkali		Immersion in deionized water	
	Maximum moisture uptake (%)	Coefficient of diffusion ($\times 10^{-7}$ mm ² /s)	Maximum moisture uptake (%)	Coefficient of diffusion ($\times 10^{-7}$ mm ² /s)
23	0.298	2.17	0.164	1.39
40	1.426	3.26	0.529	2.17
60	1.929	4.42	0.569	2.70
80	4.205	5.95	0.623	3.14

the maximum uptake and the coefficients of diffusion are higher as a result of immersion in the alkali solution. This is due to both the increased level of ester hydrolysis in the bulk resin, because of penetration of the high pH solution which induces microcracks leading to increased weight gain, and increased level of interfacial degradation, resulting in wicking along the fiber surfaces.

The activation energy for diffusion, E_d , provides a convenient indication of the energy barrier that has to be overcome for diffusion of moisture to take place and can be determined using an Arrhenius type of relationship [9]

$$D = D_0 \exp\left(\frac{-E_d}{RT}\right)$$

where D_0 is a constant coefficient, R is the universal gas constant (8.3143 J/mol K), D is the diffusion coefficient, and T is the temperature expressed on the Kelvin scale. Plotting $\ln(D)$ versus $(1/T)$ values of activation energy for the composites immersed in deionized water and alkali solution are determined to be 12.141 and 15.147 kJ/mol K, whereas the activation energy for the unreinforced resin in deionized water is 20.45 kJ/mol K. The value determined for deionized water immersion is similar to the value reported by Harper and Naeen [2] of 12.618 kJ/mol K for a unidirectional E-glass/411-45 vinylester composite, and that reported by Karbhari and Zhang [3] of 11.72 kJ/mol K for a unidirectional E-glass/411-350 vinylester composite.

3.2. Tensile response

A progression of change in tensile strength as a function of time and temperature of immersion is shown in Fig. 1(a) and (b) for the specimens immersed in deionized water and alkali solution, respectively, from which it can clearly be seen that the level of degradation increases substantially with increase in temperature of the aqueous medium. It is of significant interest to note that the degradation over the first 10 weeks is greater due to immersion in the alkaline solution, 86.3%, 68.8%, 55.9% and 46.3% retention in alkaline solution at 23, 40, 60 and 80 °C, respectively, compared to 92.4%, 70.9%, 58.8% and 48% retention in deionized water at 23, 40, 60 and 80 °C, respectively, with the difference becoming

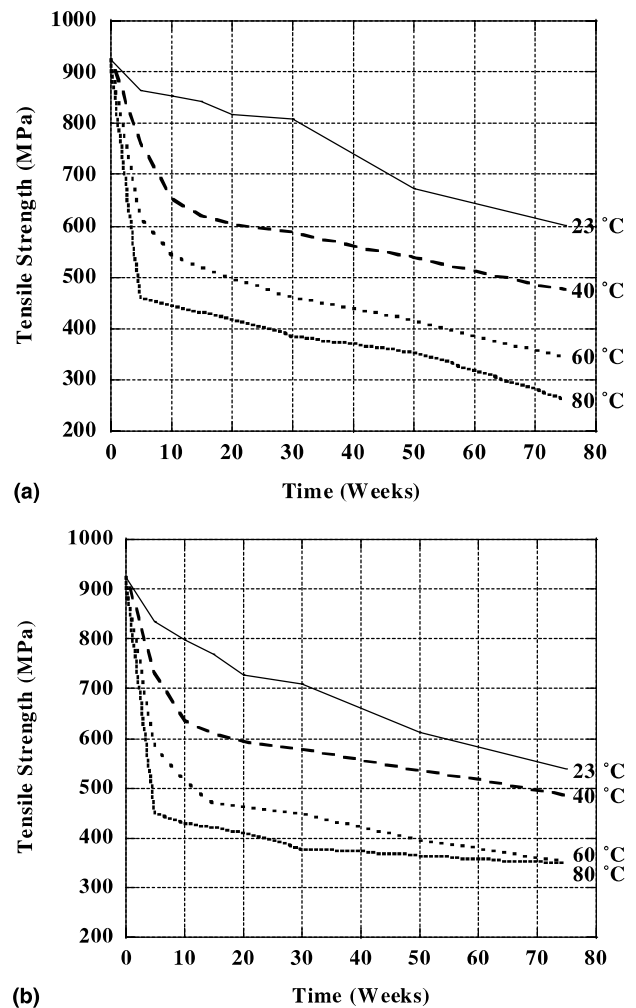


Fig. 1. (a) Change in tensile strength as a function of time and temperature of immersion in deionized water, (b) change in tensile strength as a function of time and temperature of immersion in deionized water.

smaller with increase in temperature of immersion. However, at the end of the 75-week time period the difference in values is within scatter bounds at 40 and 60 °C levels. However, at 23 °C (where the retention in strength is 58.2% in the alkaline solution and 65.2% in deionized water) the samples in alkali solution show a higher level of degradation, whereas the order is reversed at 80 °C (where the retention in strength is 37.5% in the alkaline solution and 28.2% in deionized water).

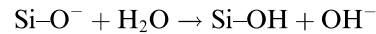
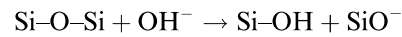
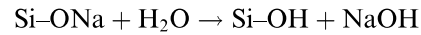
It is emphasized, however, that a mere comparison of such data is insufficient to gauge the actual degree of deterioration since performance usually is regained to an extent after the moisture absorbed is removed. In most civil infrastructure applications, with the exception of those which entail submergence of the specimen, the actual moisture content will vary based on the surrounding environment and the seasonal climate. Thus is of significant interest to actually measure regain of performance when the samples are reconditioned for the same time period and under the same conditions as their initial preconditioning. It is thus of interest to assess the level of regain due to “drying out” during reconditioning. The level of percentage regain due to reconditioning is determined as

$$\% \text{Regain} = \left[\frac{(\sigma_t)_{\text{dry}} - (\sigma_t)_{\text{wet}}}{\sigma_0 - (\sigma_t)_{\text{wet}}} \right] \times 100$$

where σ_0 is the unexposed strength, $(\sigma_t)_{\text{wet}}$ is the strength after immersion for time t in deionized water at the temperature under consideration, and $(\sigma_t)_{\text{dry}}$ is the strength of the same set but after reconditioning at 23 °C and 46% RH for 28 days (which is equal to the original period of conditioning). The denominator provides the loss in composite strength due to immersion, and the numerator provides the absolute value of regain due to reconditioning. A comparison of results is shown in Table 2 from which it can clearly be seen that there is substantially less regain (in some cases almost none) after immersion in alkaline solution indicating the exposure to these conditions results in considerable irreversible damage.

As a result of immersion in deionized water there is a relatively large loss in the first 5–10 weeks most of which is recovered on drying and is due to effects of matrix plasticization as a result of moisture uptake. As can be seen from Table 2 in the initial stages there is a significant regain of strength as a result of reconditioning (redrying) with the level of regain decreasing with both time and temperature of immersion. The higher temperatures of immersion, which can be considered as a means of acceleration of systems deterioration in time, result in a greater degree of irreversible degradation at

the level of the fiber–matrix interphase and even the fiber itself. The sorbed moisture is seen to attack the glass fiber surface with the free hydroxides that form, further degrading the silica structure. Reactions, as reported by Ishai [10] lead to the cleavage of the silicon–oxygen bonds and then the conversion to hydroxysilanes



The initiation of pitting is seen on the fiber surface due to loss of K_2O and Na_2O from the fiber by dissolution and is corroborated by the ICP results. The free hydroxyl groups are known to cause autocatalyzation of hydrolysis of ester groups in the bulk resin as well leading to a reduction in cross-link density, which is also seen by a visible change in color of the composite specimens.

Exposure to the alkali solution, however, results in significantly greater irreversible damage. Fig. 2 shows a high level of surface degradation and pitting. It is noted that glass fibers are affected by alkaline solutions through breaking of Si–O bonds in the glass network

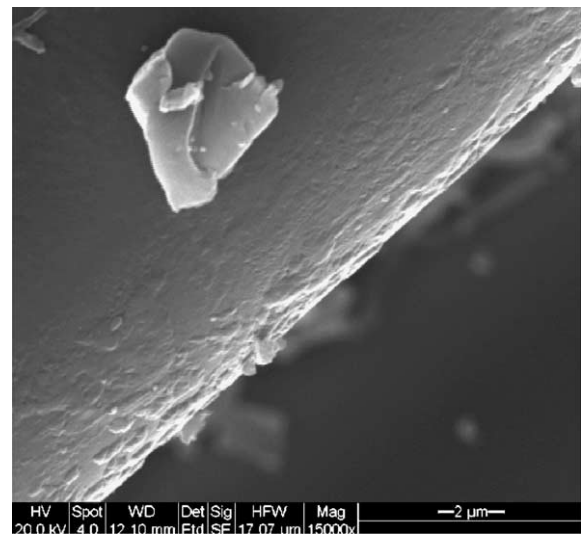


Fig. 2. Close-up showing fiber degradation at 15000× under a scanning electron microscope (SEM).

Table 2

Comparison of percentage regain in tensile strength due to reconditioning after exposure

Time period (weeks)	% Regain at 23 °C		% Regain at 40 °C		% Regain at 60 °C		% Regain at 80 °C	
	Water	Alkali	Water	Alkali	Water	Alkali	Water	Alkali
5	62.3	15.0	34.8	38.5	3.4	9.9	9.1	10.8
10	62.4	38.8	41.2	34.5	20.4	7.3	5.5	5.2
15	33.4	49.9	30.6	6.4	9.3	11.1	3.6	3.3
20	9.1	52.1	14.8	6.8	5.9	0.9	0.9	0.5
30	5.2	37.6	17.3	9.6	1.1	0.0	1.8	0.5
50	27.3	11.4	11.8	4.2	6.9	1.1	2.4	0.1
75	33.1	4.9	12.3	3.4	6.0	0.6	2.7	0.0

and this leads to both surface loss as well as deep pitting in areas of contact of high pH solution (such as concrete pore water) and $\text{Ca}(\text{OH})_2$ salts. The presence of water in itself at the fiber surface produces free alkali hydroxide groups that degrade the silica structure of the glass fiber. Energy dispersive X-ray (EDX) spectroscopy analysis of samples immersed in 60 and 80 °C alkaline solutions at times greater than 20 weeks, indicate migration of Ca^{+} towards the fiber and clustering around the fibers in the form of calcium hydroxide. Although pitting and surface roughness is seen there is no indication of spiral or longitudinal cracking of the fiber itself.

In comparison to the degradation noted in tensile strength due to immersion in water, there is only a 9.07–13.25% decrease in tensile modulus over the same time period with the decrease increasing with temperature. Regain is very high in the first 30 weeks as well. The degradation, as can be expected, is greater after alkali solution exposure with very little regain on reconditioning. This indicates that most of the damage mechanisms initiated by alkaline exposure are interface and fiber related, rather than merely at the resin level. It should be noted that the fiber level pitting at time scales longer than those considered in this study could lead to spiral and longitudinal cracking which would cause more significant drops in modulus.

3.3. Interfacial shear characterization

Short-beam-shear (SBS) tests are often used to assess interlaminar shear strength and to compare the relative effects of environmental exposure. As can be seen from Table 3 short-beam-shear strength decreases with both time of immersion and temperature of solution. In the case of immersion in deionized water (Table 3) it is noted that percentage regain at the end of the first 15 weeks of immersion are 70.62%, 57.89%, 11.33% and 10.94% for the 23, 40, 60 and 80 °C cases, respectively, clearly indicating that at higher temperatures the mechanisms of irreversible degradation were active almost right from the outset. At the end of the 75-week period of immersion there is 77.4%, 63.1%, 54.3% and 49.8% retention of SBS strength at 23, 40, 60 and 80 °C respectively, whereas the retention levels after reconditioning are 82.8%, 66%, 56.5% and 50.1%, respectively. Examination of levels of short-beam-shear strength indicate a linear relationship between SBS strength with time of immersion at 23 °C. The deterioration follows a stepped approach at 40 °C, and the more commonly observed curve having a rapid initial decrease followed by retardation is noted at 60 and 80 °C. The 23 °C results are indicative of an almost pure interlaminar failure wherein changes in strength are dependent on interfacial mechanisms only [11] without substantial physical/mechanical degradation of the matrix. The step seen in the 40 °C case is indicative of damage, primarily

Table 3
Short-beam-shear strength characteristics

Time period (weeks)	23 °C immersion temperature			40 °C immersion temperature			60 °C immersion temperature			80 °C immersion temperature		
	% Retention “wet”	% Retention “dry”	% Regain	% Retention “wet”	% Retention “dry”	% Regain	% Retention “wet”	% Retention “dry”	% Regain	% Retention “wet”	% Retention “dry”	% Regain
Deionized water: unexposed SBS strength = 43.27 Mpa												
15	95.91	98.79	70.62	96.05	98.34	57.89	83.50	85.37	11.35	70.83	74.02	10.94
30	92.00	96.14	52.02	88.88	93.46	35.39	74.21	76.29	8.06	60.27	66.44	15.53
40	90.04	94.36	43.39	85.67	88.39	19.03	68.02	69.08	3.32	59.23	60.37	2.78
50	87.13	89.76	20.47	67.69	71.02	10.30	62.33	63.72	3.68	54.29	55.21	2.02
75	77.44	82.78	23.67	63.14	66.00	7.77	54.29	56.51	4.85	49.76	50.10	0.69
Alkali solution: unexposed SBS strength = 43.27 Mpa												
15	94.01	95.22	20.08	90.02	92.08	20.60	83.27	85.86	15.47	68.25	70.77	7.93
30	91.80	92.30	6.20	81.93	82.32	2.17	72.82	73.22	1.45	60.23	61.13	2.27
40	89.23	89.74	4.72	74.46	74.79	1.27	67.92	68.22	0.94	58.05	58.59	1.27
50	85.58	85.86	1.92	67.07	67.37	0.91	59.05	59.23	0.45	53.85	54.10	0.55
75	75.11	75.2	0.37	61.13	61.20	0.18	51.26	51.47	0.43	46.55	46.71	0.30

irreversible, occurring at the interfacial level in the form of fiber–matrix debonding, and osmotic cracking at the interface, resulting in enhanced moisture sorption due to wicking. This form of damage is irreversible, and this is clearly seen in comparing the results of SBS strength of samples tested after reconditioning (redrying at 23 °C and 46% RH for 28 days) wherein an identical stepped response is seen. Beyond this level changes to the material are not merely associated with plasticization and debonding, but also to resin hydrolysis and micro-crack coalescence to form macro-transverse cracks. The latter also serve as a mechanism for enhanced wicking of moisture along the fiber. Fiber pitting and some longitudinal cracking is also seen as a result of the higher temperatures and longer periods of immersion, similar to the phenomena reported in [9,12,13].

As can be expected by the type of damage mechanisms described earlier due to immersion in alkali solution, there is a rapid initial decrease in the first 15 weeks followed by a slower reduction. The maximum drop is as a result of the severe acceleration in the 80 °C alkaline solution in which a 53.5% reduction in SBS strength is noted at the end of the 75-week period of immersion. Of this 59.4% is seen in the first 15 weeks. Similarly, specimens immersed in 23, 40 and 60 °C alkaline solutions see a 24.9%, 38.9% and 48.7% reduction, respectively, at the end of the 75-week period, of which 24%, 26% and 34%, respectively, occurs in the first 15 weeks. As noted in Table 3, regain at 15 weeks is roughly 20% for the 23 and 40 °C solutions as a result of reconditioning, with regain decreasing to 15.5% and 7.9% at the higher temperatures of 40 and 60 °C, respectively. However, after this point regain of SBS strength is extremely small, and less than that noted for tensile strength, confirming the early and aggressive attack of the alkaline solution to the interface (fiber–matrix bond) and the fiber itself, resulting in irreversible damage. Again, as with the tensile strength data, it can be seen that the level of irreversible degradation is higher due to alkaline exposure than due to immersion in deionized water.

3.4. DMTA based characterization

Since the results of moisture uptake and FTIR analysis suggest resin and interface level degradation that may not be directly characterized by mechanical tests, dynamic mechanical thermal analysis (DMTA) was used as a tool to assess changes at this level. These tests also provide for accurate assessment of thermal transitions in the material and provide distinct advantages over other forms of characterization since it is able to assess the effects of subtle physical and chemical changes in the polymer matrix, and hence also in the composite.

Water uptake by vinyl esters and their composites is known to cause plasticization in the short-term and hydrolysis over the long-term through attack of the ester

linkages. Both these phenomena induce higher levels of molecular mobility resulting in consequent decreases in the glass transition temperature, T_g , although the decrease can often be offset through residual cure of the vinyl ester itself in the aqueous solution [3,14,15]. These competing phenomena result in fluctuations in the glass transition temperature as a function of period of exposure as shown in Fig. 3. Similar trends were reported earlier by Chin et al. [16]. It is noted that the highest values of T_g , after initiation of immersion, are measured at the 10–15-week period, which coincides with the point at which the storage modulus shows a decrease after an initially significant increase. A significant increase in E' is generally indicative of progression of cure with a decrease being representative of hydrolysis and plasticization dominating effects. The progression of hydrolysis was confirmed at a preliminary level through infrared spectroscopy wherein a decrease in the ester band intensity was noted with a change in peaks at about 1452 cm^{-1} , which had been earlier noted by Ghorbel and Valentin [15] at 1450 cm^{-1} in their study on glass fiber reinforced vinyl esters.

A study of Fig. 3 shows further that the higher temperatures of immersion result in an overall lower level of decline in glass transition temperatures from the unexposed control (decreases of 8%, 8.3%, 4.9% and 5.7%, determined from values in K are seen at the 75-week level at 23, 40, 60 and 80 °C, respectively). This is due to increased leaching of hydrolyzed low molecular weight flexibilizing segments of the matrix at the higher temperatures which leads to network embrittlement, followed by further degradation through chain scission, while at lower temperatures this effect is not dominant.

Although the effect of initial post-cure can clearly be seen in the specimens immersed in deionized water, this response is not as clear in the samples immersed in alkali

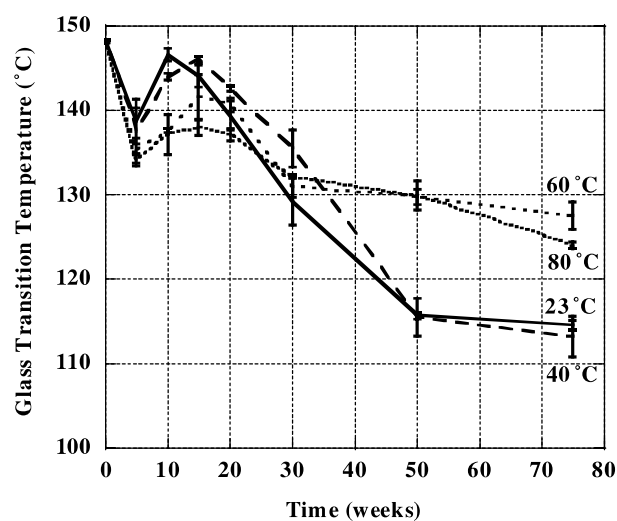


Fig. 3. Change in glass transition temperature as a function of period of immersion and temperature of deionized water.

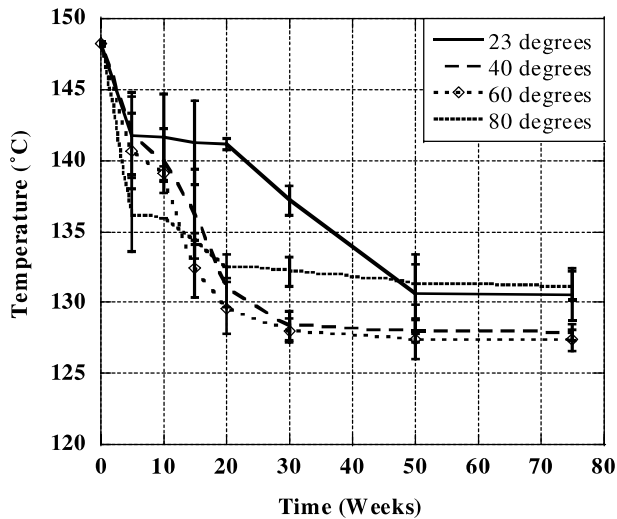


Fig. 4. Change in glass transition temperature as a function of period of immersion and temperature of alkaline solution.

solution as shown in Fig. 4. The effects of initial residual cure are seen in the initial 20 weeks in the 23 °C case wherein an initial decrease is followed by a short plateau, and then a decrease. In the 23 °C case there is a clear change from slow effects of plasticization and post-cure at the first plateau to the mode of fiber–matrix debonding, subsequent wicking of higher levels of solution, and then hydrolysis of the matrix till the second plateau. In the case of the 80 °C alkaline solution there is initially very rapid plasticization and even debonding, resulting in wicking, followed by a slower hydrolysis and deterioration. It should be noted that at this temperature moisture uptake is rapid with 1.85 wt.% being sorbed in the first 840 h, larger than the total uptake in 75 weeks at 23 and 40 °C and just short of the 1.93% sorbed at 60 °C over the 75-week period. Thus, although a linear Arrhenius fit exists between diffusion coefficients and temperature of solution (as discussed in the previous section) indicating similar mechanisms of degradation, albeit accelerated with increase in temperature, care needs to be taken in using time–temperature equivalence as a means of acceleration for predicting service life since the rates of individual degradative mechanisms appear to change significantly with temperature.

Moisture uptake in a composite can result in changes in internal stress states and as seen from Table 3 results in a decrease in the glass transition temperature. The loss tangent ($\tan \delta$), defined as the ratio of the loss modulus to the storage modulus, can be linked to the degree of molecular mobility and can hence be used as an indicator of changes in degree of cross-linking. The peak of the loss tangent occurs in the glass transition region and is hence directly related to movement of small groups and polymer chains due to transitions. The change in values of the loss tangent as a function of time

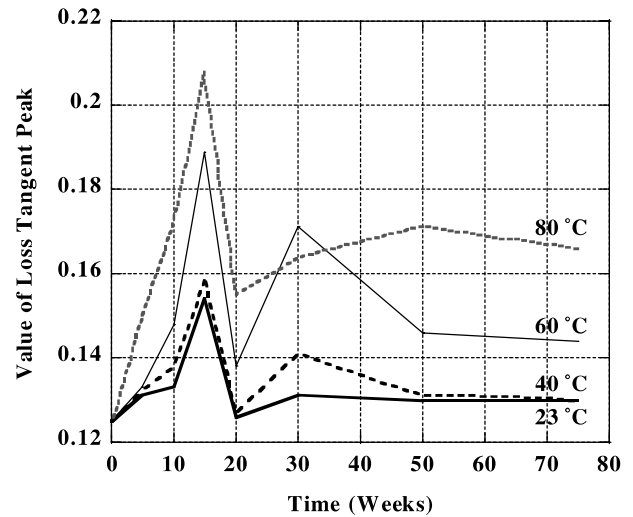


Fig. 5. Effect of time and temperature of immersion in deionized water on the peak value of the loss tangent.

of immersion and temperature of deionized water is shown in Fig. 5. It can be seen that over the first 15 weeks, which coincides with the region of highest rate of moisture gain and the time period during which degradation in mechanical properties showed reversibility, the value of the peak of the loss tangent increases. This is also in line with the previous research of Adams and Singh [17] who noted that the value of the loss factor initially increased with immersion time in seawater and then fluctuated before stabilizing at a value higher than the initial one. In addition Kuzak and Shanmugam [18] correlate the occurrence of lower values of the loss tangent and higher values of E' with increasing cross-linking and loss of ductility. Pappanicolou and Pappa [19] further found that although plasticization caused an increase in the value of the loss tangent the formation of microcavities due to accelerated deterioration through use of higher temperatures resulted in a decrease in $\tan \delta$ due to changes in morphology, which again is seen in Fig. 5 and can be correlated with irreversible damage, especially at the higher temperature levels in the form of matrix crack coalescence, microvoid formation, fiber–matrix debonding, and even fiber level degradation. A further examination of Fig. 5 shows that the value of $\tan \delta$ increases, for a given time of immersion, with the temperature of immersion, which follows earlier results reported in [19,20].

A correlation of changes in glass transition temperature with those in the loss tangent show that relative increases in T_g can be noted to occur with decreases in $\tan \delta$. The addition of styrene has been noted to increase the hydrolytic stability of vinyl esters as a result of steric shielding, but due to differing rates of polymerization in the heterogeneous resin mixture is also known to cause formation of microgels in line with the Tromsdorff effect. There is some evidence that increases in glass

transition temperature over long periods of immersion, especially at higher temperatures likely to cause rapid transition to hydrolysis and chain scission, is due to leaching of lower weight molecular species and unreacted styrene. This effect is actually seen in Fig. 3 wherein the drop in glass transition temperature is less at 60 and 80 °C, than at the 23 and 40 °C levels. Fraga et al. [21] in fact noted that in their experiments they noted an increase in T_g but a decrease in $\tan \delta$ with increase in immersion temperature of water due to extraction of styrene.

3.5. Prediction of service life

Since the diffusion data for the four temperature levels provides a linear fit to the Arrhenius equation, in the form, $\ell_n(D) = \ell_n(D_0) - E_d/(RT)$ it can reasonably be assumed, for purposes of first-order approximation, that the material follows the same mechanisms of degradation thereby allowing for use of the acceleration procedure proposed in Litherland et al. [22] and Proctor et al. [23] wherein the results of plotting the logarithm of time to attain a set of levels of normalized performance versus $1000/T$ (where T is temperature in degrees Kelvin) can be used to predict service life at a given temperature (normally taken as the ambient, or serious temperature) based on performance at higher temperatures. Using data from tests conducted on specimens immersed at 40, 60 and 80 °C predictions were made for property retention as a function of time for the 23 °C case for both tensile strength and short-beam-shear-strength and results are shown in Figs. 6 and 7, respectively.

From Fig. 6 it can be seen that over the long term the effect of alkali solution based deterioration is more than that due to deionized water. In fact the level of retention in tensile strength predicted as a result of exposure to alkali solution is almost the same as that for the specimens immersed in deionized water and then dried. It can be noted that at the end of the 50 year prediction period the specimens immersed in water show 31.15% and 47.05% retention in tensile strength in the “wet” and “dry” cases, respectively, whereas those immersed in the alkali solution showed 26.89% and 30.44% retention in the “wet” and “dry” cases, respectively.

Notwithstanding the gross simplification provided by the method of prediction, it is of interest to compare the short-term data determined through experimental means with the predictions. In the case of immersion in deionized water the results of experiments in the “wet” case provide tensile strengths in the last two periods of 50 and 75 weeks of 671.08 MPa, with a standard deviation of 23.44, and 602.2 MPa, with a standard deviation of 22.64 MPa, respectively, compared to the predicted values of 647.72 MPa at 50 weeks and 594.17 MPa at 75 weeks. Both compare well within overall scatter of experimental data. In the “dry” state the

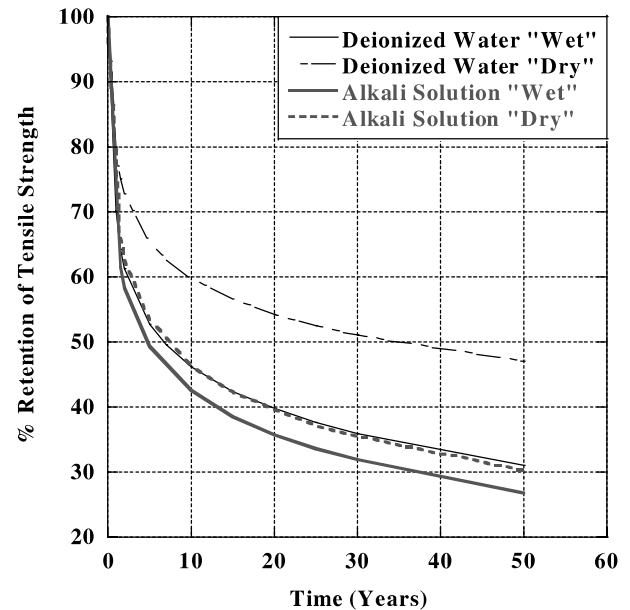


Fig. 6. Comparison of tensile strength predictions.

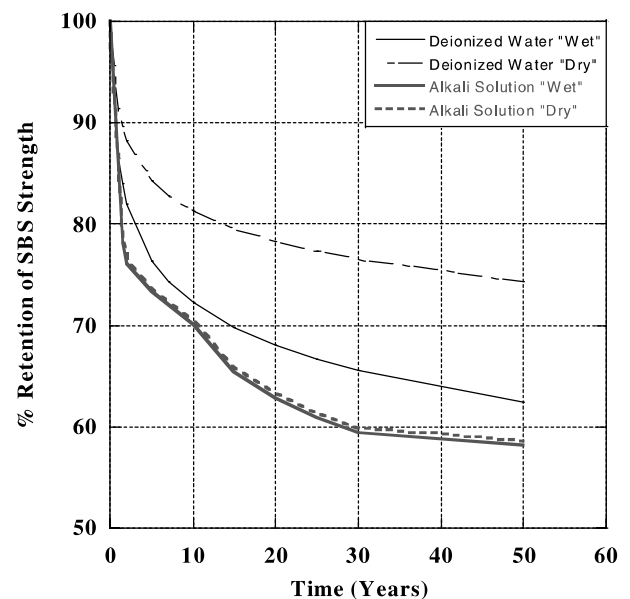


Fig. 7. Comparison of SBS strength predictions.

predictions for tensile strength at the 50- and 75-week levels are 723.07 and 693.34 MPa, respectively, which correlate well with the experimentally determined value of 739.96 MPa, with a standard deviation of 81.03 MPa, at 50 weeks, and 708.54 MPa, with a standard deviation of 43.21 MPa, at 75 weeks. In the case of the alkaline solution the experimentally determined tensile strength retention at 75 weeks (1.44 years) is 58.24% and 60.29% for the wet and dry cases, respectively, whereas the corresponding values for short beam shear strength are 75.11% and 75.20%, respectively, indicating fairly good correlation.

Prediction of short-beam-shear strength can be done following the same procedure as for the tensile strength. However, since there is clear evidence of an increase in substantial changes in mechanisms due to irreversible damage at the level of the resin and the interface, the predictions should be considered to be gross approximations at best and to be used only as a relative guide. A comparison of predictions, between the “wet” and “dry” bounds for both the deionized water and the alkali solution cases is shown in Fig. 7. It can clearly be seen that deterioration due to the exposure to alkali solution is substantially greater as well as at a higher rate through the long-term period considered. The predicted and experimental results are not as closely correlated at the longer time period of 75 weeks after immersion in deionized water in the case of SBS strength in the “dry” state due to the higher level of irreversible damage at the interfacial level which significantly affects interlaminar stress states. Experimentally determined SBS strength retention at the 50- and 75-week levels in the “wet” state are 87.13% and 82.07%, respectively, which compare well, within intrinsic data scatter bounds, with the predicted levels of 86.47% and 84.02%, respectively. In the “dry” state, however, as noted before, the predicted values of 91.34% and 89.60% retention are not as close to the experimental values of 89.76% and 82.78% retention, respectively, at the 50- and 75-week levels. Similar trends are noted in the case of alkaline solution immersion with the deviation being even greater. The larger differences as well as the non-conservative predictions can be attributed to the significant changes in degradation mechanisms in the case of interlaminar shear which is resin and interface dominated. This emphasizes the approximate nature of the predictive method and the clear indication that a direct relationship cannot always be made between progression of moisture uptake (which is the basis for calculation of activation energy) and acceleration of deterioration in performance. In the current case the predictions are close in the fiber dominated mode (tensile strength) but not in the interface and resin dominated mode (SBS strength).

The reader is cautioned that although the values based on the physical mechanisms studied the model does not consider effects of zero-stress ageing on glass fibers, nor the effects of sustained stress, both of which can be expected to affect longevity and strength retention.

4. Summary

Immersion in deionized water and alkaline solution is shown to result in a range of mechanisms including plasticization, hydrolysis, fiber–matrix debonding, ma-

trix microcracking, and microcrack coalescence. While the first of these mechanisms causes reversible decreases in mechanical properties, the others cause changes that are irreversible even on redrying of the specimens. The changes in mechanical properties are seen to be more severe due to the effect of the alkaline solution, which not only causes degradation at the level of the resin and interphase, but also that of the fiber through significant pitting and even material loss. It is seen that characteristics obtained from dynamic mechanical thermal analysis correlate well with mechanisms of damage and transitions in materials response as a result of competing mechanisms that can be traced to the heterogeneity and structure of the resin itself and the cure regime. This area of investigation is seen as a promising one for the further development of not only a mechanistic understanding of the deterioration phenomena, but also for the development of predictive models that correlate changes in sensitive parameters such as the storage and loss moduli, loss tangent, and glass transition temperature with the amount of moisture uptake, and from this to the more commonly used mechanical properties such as tensile and short-beam-shear strength.

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